Introduction to Cryptography

Lectures 6–7: Cryptography Fundamentals, Terminology, Cryptanalysis and Brute Force Attack

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Why Cryptography?

Cryptography is used when information must be protected during storage or transmission.

- Protects confidential information from unauthorized disclosure.
- Detects or prevents unauthorized modification.
- Helps verify identity and message origin.
- Supports secure communication over insecure networks.
- Provides mathematical foundations for modern cybersecurity.

Examples

Online banking, digital payments, WhatsApp messages, HTTPS websites, VPNs, cloud storage, blockchain and digital signatures.

Basic Terminology

Term	Meaning
Plaintext	Original readable message or data.
Ciphertext	Coded or scrambled message.
Encryption / Enciphering	Process of converting plaintext into ciphertext.
Decryption / Deciphering	Process of converting ciphertext back into plaintext.
Cipher	Cryptographic algorithm or system.

Plaintext and Ciphertext

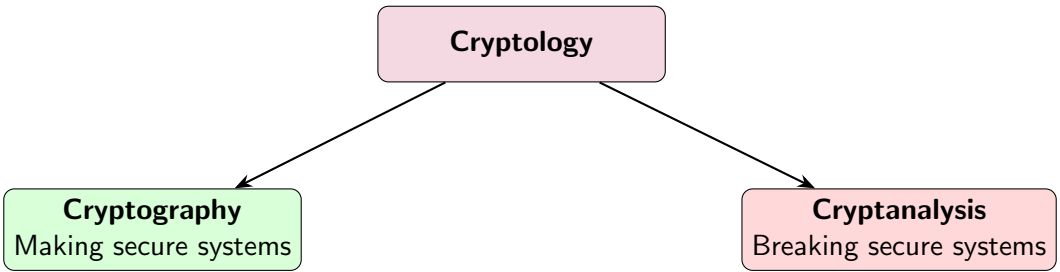


Example

Plaintext: `ATTACK AT DAWN`

Ciphertext: `DWWDFN DW GDZQ` using a simple Caesar shift of 3.

Cryptography, Cryptanalysis and Cryptology



Key Idea

Cryptology is the complete field that includes both cryptography and cryptanalysis.

Symmetric Encryption

Definition

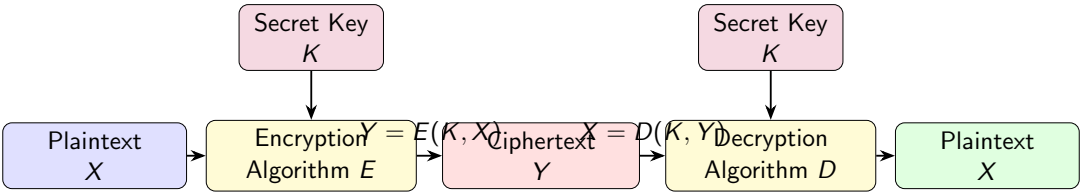
Symmetric encryption uses the same secret key for encryption and decryption.

- Also called conventional encryption, secret-key encryption or single-key encryption.
- Sender and receiver must share the same key securely.
- The algorithm may be public, but the key must remain secret.
- It is widely used because it is fast and efficient.

Examples

AES, DES, Triple DES, ChaCha20 and classical substitution ciphers.

Simplified Symmetric Encryption Model



Five Ingredients of Symmetric Encryption

A symmetric encryption scheme has five ingredients:

1. **Plaintext:** Original intelligible message.
2. **Encryption algorithm:** Performs substitutions and transformations.
3. **Secret key:** Controls the transformation performed by the algorithm.
4. **Ciphertext:** Scrambled output produced by encryption.
5. **Decryption algorithm:** Converts ciphertext back into plaintext using the same key.

Important

For the same plaintext, different keys produce different ciphertexts.

Encryption and Decryption Equations

Let:

- X = Plaintext
- Y = Ciphertext
- K = Secret key
- E = Encryption algorithm
- D = Decryption algorithm

Encryption

$$Y = E(K, X)$$

Decryption

$$X = D(K, Y)$$

Role of the Secret Key

The secret key is independent of both plaintext and algorithm.

- The same algorithm can produce different ciphertexts using different keys.
- If the key is compromised, encrypted communication becomes readable.
- Key secrecy is more important than algorithm secrecy.
- Secure key generation, distribution and storage are essential.

Example

In AES, the algorithm is public, but the 128-bit, 192-bit or 256-bit key must remain secret.

Two Requirements for Secure Symmetric Encryption

1. **Strong encryption algorithm**
   - The opponent may know the algorithm.
   - The opponent may possess ciphertexts.
   - The opponent should still be unable to recover plaintext or key.
2. **Secure key sharing and key protection**
   - Sender and receiver must obtain the key securely.
   - Both must keep the key secret.
   - If the key is discovered, all communication using it is compromised.

Kerckhoffs's Principle

Principle

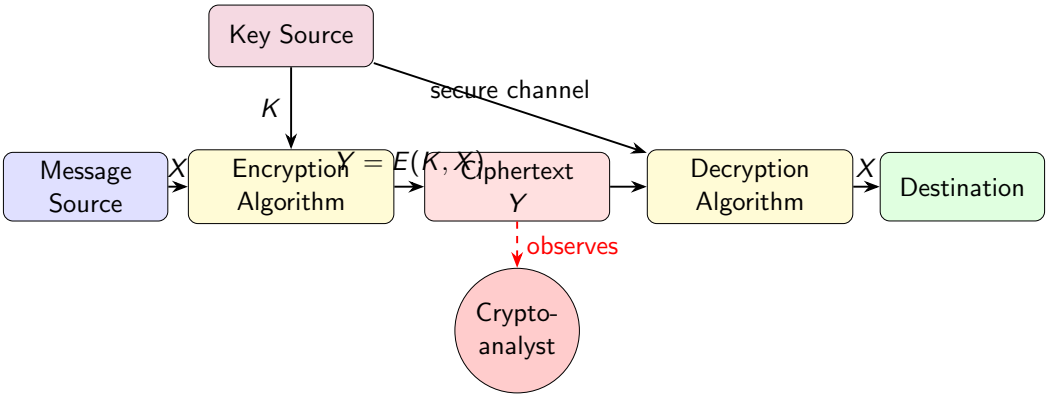
A cryptographic system should remain secure even if everything about the system is public, except the secret key.

- Security should not depend on hiding the algorithm.
- Public algorithms can be reviewed and tested by experts.
- Practical security depends mainly on key secrecy.
- Modern algorithms such as AES are publicly known.

Key Message

Do not rely on secret algorithms. Rely on strong algorithms and protected keys.

Model of Symmetric Cryptosystem



What Does the Opponent Know?

In modern cryptography, we usually assume that the opponent knows:

- The encryption algorithm E
- The decryption algorithm D
- One or more ciphertexts Y
- Sometimes some plaintext–ciphertext pairs

The opponent usually does **not** know:

- The secret key K
- The complete plaintext X

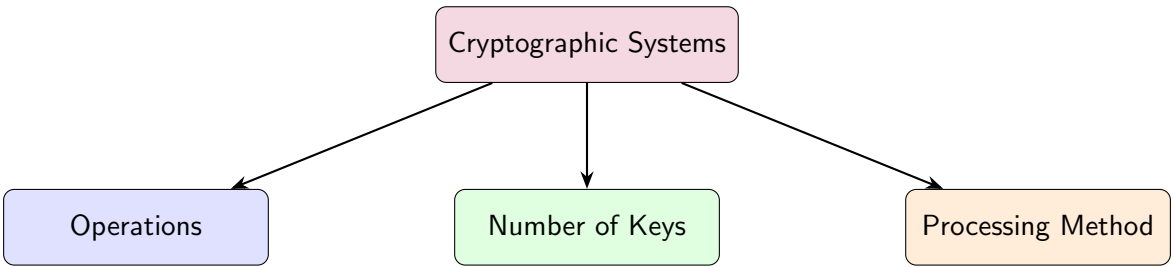
Attacker's Objective

Recover the plaintext X, the key K, or both.

Classification of Cryptographic Systems

Cryptographic systems are characterized along three independent dimensions:

1. Type of operations used for transformation
2. Number of keys used
3. Way in which plaintext is processed



Dimension 1: Type of Operations

All encryption algorithms are based on two general principles:

Substitution

Each element in the plaintext is replaced by another element.

Example

$A \rightarrow D$, $B \rightarrow E$, $C \rightarrow F$

Transposition

Elements in the plaintext are rearranged.

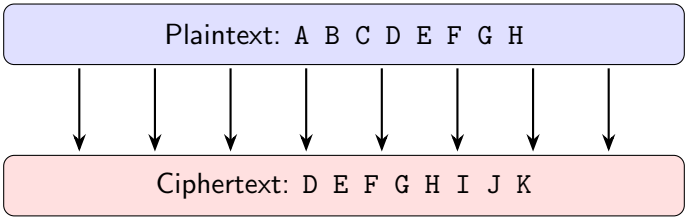
Example

`HELLO` $\rightarrow$ `OLELH`

Requirement

No information should be lost. The operation must be reversible.

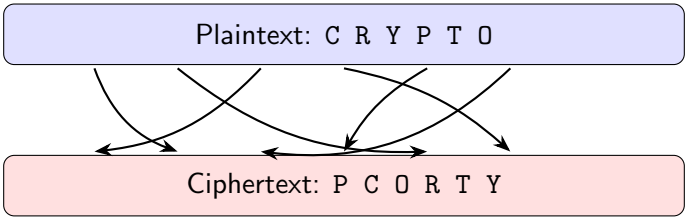
Substitution Cipher: Simple Illustration



Observation

The position of symbols remains the same, but each symbol is replaced.

Transposition Cipher: Simple Illustration



Observation

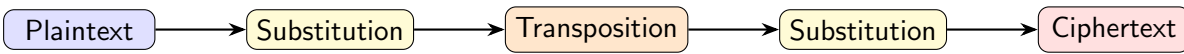
The symbols are not replaced. Their positions are rearranged.

Product Systems

Definition

A product system combines multiple stages of substitution and transposition.

- Most modern ciphers use repeated rounds.
- Each round may include substitution, permutation and key mixing.
- Repetition increases confusion and diffusion.
- AES is an example of a modern product cipher.



Dimension 2: Number of Keys

Symmetric Encryption

- Same key for encryption and decryption.
- Also called secret-key encryption.
- Fast and efficient.
- Key distribution is difficult.

Asymmetric Encryption

- Uses two different keys.
- Public key and private key.
- Easier key distribution.
- Generally slower than symmetric encryption.

Symmetric vs Asymmetric Encryption

Feature	**Symmetric**	**Asymmetric**
Keys used	One shared secret key	Public key and private key
Speed	Faster	Slower
Key distribution	Difficult	Easier
Main use	Bulk data encryption	Key exchange, signatures
Examples	AES, DES	RSA, ECC

Dimension 3: Processing Method

Block Cipher

Processes plaintext one block at a time and produces one output block for each input block.

- Fixed block size
- Example: AES uses 128-bit blocks

Stream Cipher

Processes input elements continuously and produces output one element at a time.

- Bit-by-bit or byte-by-byte processing
- Example: ChaCha20

Block Cipher vs Stream Cipher

Feature	**Block Cipher**	**Stream Cipher**
Input processing	Fixed-size blocks	Continuous stream
Common unit	Block of bits	Bit or byte
Error impact	May affect block/mode	Usually local impact
Examples	AES, DES	RC4, ChaCha20
Use case	Files, databases, TLS modes	Real-time communication

Classroom Example: Encrypting a Message

Suppose Alice wants to send Bob the message:

Plaintext

```
MEET AT 10 AM
```

Alice and Bob share a secret key:

Secret Key

```
K = 7
```

Using a simple shift cipher:

- Alice shifts each letter by 7 to produce ciphertext.
- Bob shifts each letter backward by 7 to recover plaintext.
- Anyone without the key should not easily read the message.

Important Terminology Summary

Term	**Short Meaning**
Plaintext	Original readable data
Ciphertext	Encrypted unreadable data
Encryption	Plaintext to ciphertext
Decryption	Ciphertext to plaintext
Key	Secret value controlling encryption/decryption
Cipher	Cryptographic algorithm/system
Cryptography	Designing secure communication techniques
Cryptanalysis	Breaking cryptographic systems
Cryptology	Cryptography + cryptanalysis

What is Cryptanalysis?

Definition

Cryptanalysis is the study of techniques for deciphering a message without knowing the complete enciphering details or the secret key.

- Commonly called "breaking the code."
- Attempts to exploit weaknesses in algorithms, implementation or usage.
- May use statistical patterns of plaintext and ciphertext.
- Often aims to recover the secret key, not just one plaintext.

Goal of Attacking an Encryption System

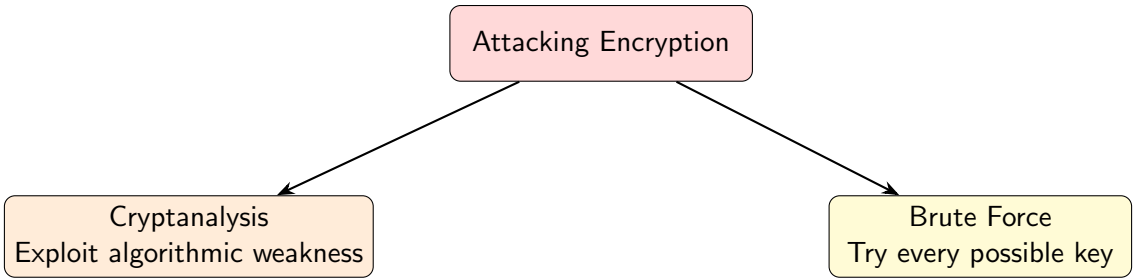
The attacker may attempt to recover:

1. The plaintext of a specific ciphertext.
2. The secret key used for encryption.
3. Future messages encrypted with the same key.
4. Past messages encrypted with the same key.

Catastrophic Impact

If the secret key is recovered, all past and future messages encrypted with that key may be compromised.

Two General Approaches to Attacking Encryption



Cryptanalytic Attack

Cryptanalytic attacks rely on:

- Knowledge of the encryption algorithm
- General characteristics of plaintext
- Statistical properties of language or data
- Sample plaintext–ciphertext pairs
- Structural weaknesses in the cipher

Example

If a plaintext is English text, letters such as E, T, A and O may occur frequently. A weak cipher may preserve such patterns.

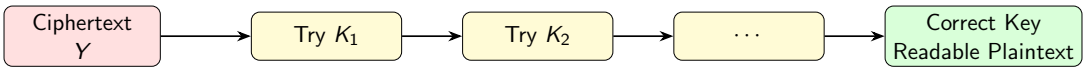
Brute-Force Attack

Definition

A brute-force attack tries every possible key on a ciphertext until an intelligible plaintext is obtained.

- Does not require deep knowledge of cipher structure.
- Depends heavily on key-space size.
- On average, half of all possible keys must be tried.
- If the key length is large, brute force becomes computationally infeasible.

Brute-Force Attack: Illustration



Average Work

For a key space of size 2^n, average number of trials is 2^{n-1}.

Key Space

Definition

Key space is the total number of possible keys that can be used in a cryptographic system.

If the key length is n bits:

$$\text{Key Space} = 2^n$$

Average brute-force attempts:

$$\text{Average Attempts} = 2^{n-1}$$

Example

A 10-bit key has $2^{10} = 1024$ possible keys. On average, about 512 keys must be tried.

Effect of Key Size

Key Size	Total Keys	Average Trials
8-bit	$2^8 = 256$	$2^7 = 128$
16-bit	$2^{16} = 65,536$	2^{15}
32-bit	2^{32}	2^{31}
56-bit	2^{56}	2^{55}
128-bit	2^{128}	2^{127}

Key Idea

Increasing the key size by 1 bit doubles the brute-force search space.

Types of Cryptanalytic Attacks

The strength of an attack depends on how much information is available to the cryptanalyst.

1. Ciphertext-only attack
2. Known-plaintext attack
3. Chosen-plaintext attack
4. Chosen-ciphertext attack
5. Chosen-text attack

Observation

The more information the attacker has, the stronger the attack model becomes.

Ciphertext-Only Attack

Known to the Cryptanalyst

- Encryption algorithm
- Ciphertext

- This is the most difficult case for the attacker.
- The attacker may use brute force or statistical analysis.
- Weak classical ciphers may fail under this attack.

Example

An attacker intercepts only encrypted messages traveling over a network.

Known-Plaintext Attack

Known to the Cryptanalyst

- Encryption algorithm
- Ciphertext
- One or more plaintext–ciphertext pairs produced using the secret key

- More powerful than ciphertext-only attack.
- Common when message formats have predictable headers.
- The attacker studies how known plaintext is transformed.

Known-Plaintext Example

Some files and protocols contain predictable data.

- File headers
- Email templates
- Payment transaction formats
- Standard greetings
- Copyright statements in source code

Example

If every file begins with a known header, the attacker may know part of the plaintext and the matching ciphertext.

Probable-Word Attack

Idea

A probable-word attack guesses that certain words or patterns are likely to appear in the plaintext.

Examples of probable words:

- `CONFIDENTIAL`
- `TRANSFER`
- `PASSWORD`
- `AMOUNT`
- `DATE`
- Company copyright notice

Connection

Probable-word attacks are closely related to known-plaintext attacks.

Chosen-Plaintext Attack

Known to the Cryptanalyst

- Encryption algorithm
- Ciphertext
- Plaintext messages chosen by the cryptanalyst
- Corresponding ciphertext generated using the secret key

- The attacker can choose inputs and observe encrypted outputs.
- Carefully chosen patterns may reveal cipher structure.
- Differential cryptanalysis is an example strategy.

Chosen-Ciphertext Attack

Known to the Cryptanalyst

- Encryption algorithm
- Ciphertext
- Ciphertexts chosen by the cryptanalyst
- Corresponding decrypted plaintexts

- The attacker obtains decryptions of selected ciphertexts.
- This is important in protocol and public-key security analysis.
- Secure systems must prevent misuse of decryption oracles.

Chosen-Text Attack

Known to the Cryptanalyst

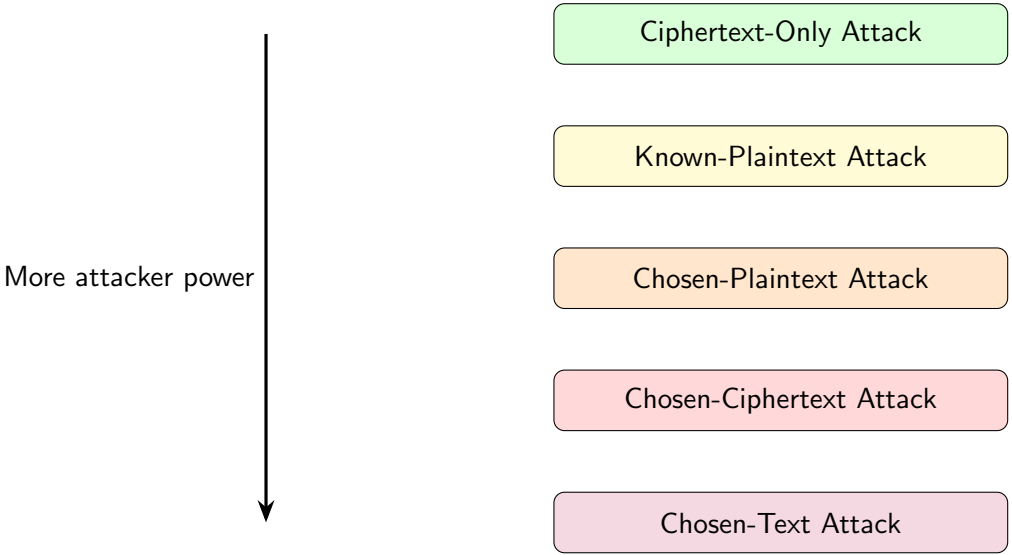
The cryptanalyst can choose plaintext messages and ciphertext messages and obtain corresponding outputs.

- Combines chosen-plaintext and chosen-ciphertext capabilities.
- Strong attack model.
- Used mainly for theoretical security analysis.

Comparison of Attack Types

Attack Type	**Information Available to Cryptanalyst**
Ciphertext-only	Encryption algorithm and ciphertext
Known-plaintext	Ciphertext and one or more plaintext–ciphertext pairs
Chosen-plaintext	Plaintexts chosen by attacker and corresponding ciphertexts
Chosen-ciphertext	Ciphertexts chosen by attacker and corresponding plaintexts
Chosen-text	Both chosen plaintext and chosen ciphertext capability

Attack Strength Ladder



Unconditional Security

Definition

An encryption scheme is unconditionally secure if the ciphertext does not contain enough information to uniquely determine the plaintext, no matter how much ciphertext or time the opponent has.

- Security is information-theoretic.
- Even unlimited computing power cannot break it.
- The one-time pad is the classic example.
- Most practical systems are not unconditionally secure.

Computational Security

Definition

An encryption scheme is computationally secure if breaking it is computationally infeasible with available resources.

A cipher is considered computationally secure if:

1. The cost of breaking the cipher exceeds the value of the encrypted information.
2. The time required to break the cipher exceeds the useful lifetime of the information.

Practical Security

Most modern cryptographic systems aim for computational security.

Unconditional vs Computational Security

Feature	**Unconditional Security**	**Computational Security**
Attacker power	Unlimited	Limited by resources
Security basis	Information theory	Computational difficulty
Breakability	Impossible in principle	Possible but impractical
Example	One-time pad	AES, RSA, ECC
Practicality	Difficult key management	Widely used

Why Key Length Matters

A brute-force attacker tries keys from the key space.

- A larger key means a larger search space.
- Every extra bit doubles the number of possible keys.
- Small keys can be broken quickly with modern hardware.
- Very large keys make brute-force attacks impractical.

Formula

For an n-bit key, total possible keys are 2^n.

Key Size Examples

Algorithm / Case	Key Size	Security Observation
DES	56-bit	No longer secure against modern brute force
Triple DES	168-bit nominal	Stronger than DES, but largely legacy
AES-128	128-bit	Currently considered strong for general use
AES-256	256-bit	Very large key space

Important

Brute-force feasibility depends on both key length and attacker computing power.

Simple Brute-Force Time Calculation

Suppose a machine can test 10^6 keys per second.

For a 40-bit key

$$2^{40} \approx 1.1 \times 10^{12} \text{ keys}$$

$$\text{Average trials} \approx 2^{39}$$

- Higher testing rates reduce attack time.
- Parallel machines can test many keys simultaneously.
- This is why modern systems require large key sizes.

Why DES Became Insecure

- DES uses a 56-bit key.
- 56-bit key space was large when DES was designed.
- Modern hardware can search this space much faster.
- Massively parallel systems make brute force more practical.
- Therefore, DES is no longer considered computationally secure.

Lesson

A key size that is secure today may become insecure as computing power increases.

Patterns and Cryptanalysis

Many cryptanalytic attacks exploit the fact that plaintext has structure.

- Natural languages have frequency patterns.
- Files have headers and standard formats.
- Protocol messages have predictable fields.
- Source code may contain repeated keywords.
- Financial messages may contain fixed templates.

Good Encryption

A strong cipher should hide plaintext structure so that ciphertext appears random.

Attacker Knowledge in Practice

In practical attacks, the attacker may know:

- The protocol being used
- File format or message format
- Common words or standard headers
- Approximate timing or size of messages
- Some plaintext–ciphertext pairs
- Implementation weaknesses

Security Engineering Rule

Design systems to remain secure even under realistic attacker knowledge.

Cryptanalysis vs Brute Force

Feature	**Cryptanalysis**	**Brute Force**
Method	Exploit weakness or patterns	Try every possible key
Requires algorithm insight	Yes	Not necessarily
Main dependency	Cipher structure and known data	Key-space size
Result	Plaintext or key recovery	Key recovery
Defence	Strong cipher design	Large key size

Real-World Scenario: Online Banking

Consider an encrypted online banking transaction.

- **Plaintext:** Account number, amount and beneficiary.
- **Ciphertext:** Encrypted transaction data.
- **Key:** Session key negotiated during secure connection.
- **Cryptanalysis risk:** Weak cipher or predictable message format.
- **Brute-force risk:** Small key size.
- **Defence:** Strong algorithms, long keys, secure protocols and key rotation.

Class Activity 1

Identify the term:

1. Original readable data.
2. Encrypted unreadable data.
3. Process of converting plaintext into ciphertext.
4. Process of recovering plaintext from ciphertext.
5. Study of breaking cryptographic systems.
6. Total field containing cryptography and cryptanalysis.

Class Activity 1: Answers

1. Plaintext
2. Ciphertext
3. Encryption / Enciphering
4. Decryption / Deciphering
5. Cryptanalysis
6. Cryptology

Class Activity 2

For each case, identify the attack type:

1. The attacker has only encrypted messages.
2. The attacker has some plaintext messages and their ciphertexts.
3. The attacker can choose plaintexts and obtain ciphertexts.
4. The attacker can choose ciphertexts and obtain decrypted plaintexts.
5. The attacker tries all possible keys one by one.

Class Activity 2: Answers

1. Ciphertext-only attack
2. Known-plaintext attack
3. Chosen-plaintext attack
4. Chosen-ciphertext attack
5. Brute-force attack

Quick Quiz

1. Define plaintext and ciphertext.
2. Differentiate between cryptography and cryptanalysis.
3. Draw the simplified symmetric encryption model.
4. List the five ingredients of symmetric encryption.
5. State the two requirements for secure symmetric encryption.
6. Explain substitution and transposition.
7. Differentiate between block cipher and stream cipher.
8. What is brute-force attack?
9. What is known-plaintext attack?
10. Define computational security.

References

- William Stallings, *Cryptography and Network Security: Principles and Practice*.
- William Stallings and Lawrie Brown, *Computer Security: Principles and Practice*.
- NIST, *Cryptographic Standards and Guidelines*.
- RFC 4949, *Internet Security Glossary, Version 2*.

Thank You

Questions and Discussion